



MODUL **PRAKTIKUM**

ADMINISTRASI BASIS DATA

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Materi 9

PEMELIHARAAN DATABASE (MAINTENANCE)

A. Tujuan

Setelah mempelajari materi ini, mahasiswa diharapkan mampu:

- Mengidentifikasi tingkat fragmentasi index
- Membuat simulasi fragmentasi secara sengaja
- Menentukan tindakan REORGANIZE atau REBUILD
- Melakukan update statistics
- Menggunakan DBCC CHECKDB untuk integritas database
- Menganalisis perubahan sebelum dan sesudah maintenance

B. Persiapan

- Gunakan SQL Server Management Studio.
- Pastikan memiliki hak akses penuh pada database.

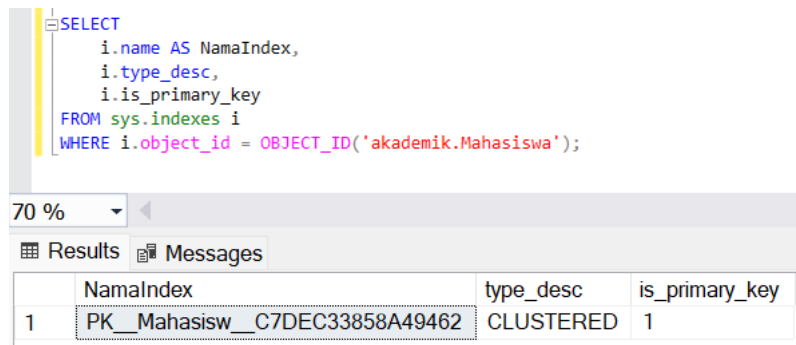
C. Praktikum

1 - IDENTIFIKASI INDEX EXISTING

Langkah 1 — Gunakan Database

```
USE DBKampus;
```

Langkah 2 – Cek Struktur Index¹



The screenshot shows the SQL Server Management Studio interface. The query editor contains the following SQL code:

```
SELECT  
    i.name AS NamaIndex,  
    i.type_desc,  
    i.is_primary_key  
FROM sys.indexes i  
WHERE i.object_id = OBJECT_ID('akademik.Mahasiswa');
```

Below the query editor, the 'Results' tab is active, displaying a table with the following data:

	NamaIndex	type_desc	is_primary_key
1	PK_Mahasisw__C7DEC33858A49462	CLUSTERED	1

Mahasiswa akan melihat:

- PK_Mahasiswa (Clustered Index pada NIM)

2 – CEK FRAGMENTASI AWAL²

```
SELECT
    OBJECT_NAME(object_id) AS>NamaTabel,
    avg_fragmentation_in_percent,
    page_count
FROM sys.dm_db_index_physical_stats
(DB_ID(), OBJECT_ID('akademik.Mahasiswa'), NULL, NULL, NULL);
```

70 %

 Results  Messages

	NamaTabel	avg_fragmentation_in_percent	page_count
1	Mahasiswa	0	1

Biasanya hasil masih 0–5%.

3 – MEMBUAT FRAGMENTASI (5-30%)

Karena NIM bertipe char(11) dan menjadi clustered index, kita akan memasukkan data acak agar terjadi page split.

Langkah 1 – Backup Data Awal (Opsional Aman)

```
SELECT * INTO akademik.Mahasiswa_backup
FROM akademik.Mahasiswa;
```

70 %

Messages

(6 rows affected)

Completion time: 2026-05-12T09:24:18.2276388+07:00

Langkah 2 – Insert Data Acak³

[illegible]

Langkah 3 – Cek Fragmentasi Lagi

```
SELECT
    avg_fragmentation_in_percent,
    page_count
FROM sys.dm_db_index_physical_stats
(DB_ID(), OBJECT_ID('akademik.Mahasiswa'), NULL, NULL, NULL);
```

70 %

Results Messages

	avg_fragmentation_in_percent	page_count
1	92.5925925925926	27

Target: 5–30% atau lebih, Jika belum mencapai 5%, ulangi insert.

4 – REORGANIZE (Untuk 5–30% atau lebih)⁴

```
ALTER INDEX ALL ON akademik.Mahasiswa
REORGANIZE;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-05-12T09:38:05.6605888+07:00

Cek kembali fragmentasi → harus turun.

```
SELECT
    avg_fragmentation_in_percent,
    page_count
FROM sys.dm_db_index_physical_stats
(DB_ID(), OBJECT_ID('akademik.Mahasiswa'), NULL, NULL, NULL);
```

70 %

Results Messages

	avg_fragmentation_in_percent	page_count
1	5.55555555555556	18

5 – REBUILD⁵

```
ALTER INDEX ALL ON akademik.Mahasiswa
REBUILD;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-05-12T09:46:00.6706552+07:00

Cek kembali fragmentasi → turun mendekati 0%.

```
SELECT
    avg_fragmentation_in_percent,
    page_count
FROM sys.dm_db_index_physical_stats
(DB_ID(), OBJECT_ID('akademik.Mahasiswa'), NULL, NULL, NULL);
```

70 %

Results Messages

	avg_fragmentation_in_percent	page_count
1	0	18

6 – UPDATE STATISTICS

Cek Statistik

- Cari Nama Index / Statistik

```
SELECT name
FROM sys.stats
WHERE object_id = OBJECT_ID('akademik.Mahasiswa');
```

70 %

Results Messages

	name
1	PK_Mahasisw_C7DEC33858A49462

- Gunakan Nama Tersebut⁶

```
DBCC SHOW_STATISTICS ('akademik.Mahasiswa', 'PK_Mahasisw_C7DEC33858A49462');
```

70 %

Results Messages

	Name	Updated	Rows	Rows Sampled	Steps	Density	Average key length	String Index	Filter Expression	Unfiltered Rows
1	PK_Mahasisw_C7DEC33858A49462	May 12 2026 9:46AM	3006	3006	152	1	11	YES	NULL	3006

	All density	Average Length	Columns
1	0.000332668	11	NIM

	RANGE_HI_KEY	RANGE_ROWS	EQ_ROWS	DISTINCT_RANGE_ROWS	AVG_RANGE_ROWS
1	00001199452	0	1	0	1
2	00010624519	14	1	14	1
3	00029191471	19	1	19	1
4	00035631841	11	1	11	1
5	00048305711	19	1	19	1
6	00057027279	14	1	14	1
7	00061616287	7	1	7	1

Update Statistik

```
UPDATE STATISTICS akademik.Mahasiswa;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-05-12T09:54:02.1962427+07:00

Atau seluruh database:

```
EXEC sp_updatestats;
```

70 %

Messages

Updating [akademik].[Mahasiswa_backup]
0 index(es)/statistic(s) have been updated, 0 did not require update.

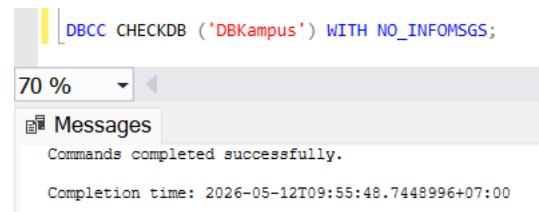
Updating [sys].[sqlagent_jobs]
[sqlagent_jobs_clust], update is not necessary...
[sqlagent_jobs_nc1_name], update is not necessary...
0 index(es)/statistic(s) have been updated, 2 did not require update.

Updating [sys].[sqlagent_jobsteps]
[sqlagent_jobsteps_clust], update is not necessary...
[sqlagent_jobsteps_nc1], update is not necessary...
[sqlagent_jobsteps_nc2], update is not necessary...
0 index(es)/statistic(s) have been updated, 3 did not require update.

Updating [sys].[sqlagent_job_history]
[sqlagent_job_history_clust], update is not necessary...
[sqlagent_job_history_nc1], update is not necessary...
0 index(es)/statistic(s) have been updated, 2 did not require update.

Updating [sys].[sqlagent_jobsteps_logs]

7 – DBCC CHECKDB⁷



Jika tidak ada error → database sehat.